

LOW ACRYLONITRILE NBR PATENT LANDSCAPE: COMPOSITION CONTROL, DEGRADATION, AND SWELLABLE RUBBER APPLICATIONS

1. ABSTRACT

This report synthesizes two primary patents relevant to low acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile-rubber processing. The target material is defined as NBR with reduced acrylonitrile fraction relative to conventional grades, typically pursued to improve low-temperature flexibility while retaining sufficient oil resistance. The patent landscape here is not centered on direct low-ACN copolymer synthesis; instead, it spans downstream molecular-weight reduction of nitrile rubber and formulation of nitrile-elastomer-containing swellable rubber compositions. EP2145681B1 addresses metathesis-based molecular-weight reduction of nitrile rubber using ruthenium or osmium metathesis catalysts combined with a boron-trifluoride adduct activator. The disclosure is relevant because it enables lower-Mooney, more processable nitrile rubber feedstocks, which can be useful when tailoring NBR grades for improved handling or for subsequent hydrogenation/degradation workflows. US10385647B2 focuses on swellable rubber compositions containing a nitrile elastomer, butadiene rubber, sulfur curing agent, silica, and powder super absorbent polymer (SAP), with explicit embodiments using nitrile rubber elastomers and hydrogenated nitrile elastomers. Although not a synthesis patent for low-ACN NBR, it provides formulation/process guidance showing how nitrile elastomers are incorporated into high-temperature, high-pressure swellable packers and how composition, SAP loading, and curing conditions affect performance. Across the two patents, the major trends are: use of metathesis chemistry to reduce molecular weight and improve processability; use of nitrile elastomer blends in demanding oilfield swellable systems; and strong emphasis on temperature, pressure, and swelling-performance windows rather than direct control of acrylonitrile content. Overall, the evidence suggests that low-ACN NBR synthesis itself is not directly disclosed in these primary patents, but the patents provide adjacent technical routes for modifying nitrile rubber properties and for deploying nitrile elastomers in formulations where low-temperature flexibility, swelling behavior, and processability are critical.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2145681B1
- Patent Title: Method for the reduction of the molecular weight of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a metathesis-based process for reducing the molecular weight of nitrile rubber by contacting NBR with a catalyst system and reacting it. The catalyst system includes a metathesis catalyst, typically a ruthenium- or osmium-based complex, together with a boron-trifluoride adduct activator of formula (Z). - Hydrogenation degree of HNBR: 50 to 100% — background description - Mooney viscosity (commercial HNBR): 55 to 120 ML 1+4 at 100°C — background description - Number-average molecular weight (commercial HNBR): about 200,000 to 700,000 — background description - Residual double bond content: 0 to 18% — background description - Hydrogenation degree of diene units: 50 to 100% — background description - Molecular weight after degradation in cited prior art: Mw 30,000 to 250,000 — cited prior art discussion - Mooney viscosity after degradation in cited prior art: 3 to 50 ML 1+4 at 100°C — cited prior art discussion - PDI after degradation in cited prior art: less than 2.5 — cited prior art discussion - Grubbs-I catalyst loading in cited prior art: 0.5 phr — cited prior art discussion - Ru content in cited prior art: 61 ppm Ru based on nitrile rubber used — cited prior art discussion

Relevant Experimental Evidence - Example 1: No structured experimental example was detected in the provided extraction; the patent text is primarily a process disclosure and background discussion. - Example 2: The disclosure emphasizes metathesis catalyst systems for nitrile-rubber molecular-weight reduction rather than a composition example. - Example 3: The patent frames NBR as a copolymer/terpolymer of an α,β -unsaturated nitrile and a conjugated diene, supporting relevance to nitrile-rubber modification. - Example 4: The text highlights that lower molecular weight improves processability, which is the central technical objective of the process. - Example 5: The patent discusses prior-art metathesis degradation followed by hydrogenation as context for the claimed process.

Technical Relevance This patent is relevant because it provides a route to lower molecular weight nitrile rubber, which can improve processability of NBR-derived materials. While it does not directly teach low-acrylonitrile copolymerization, it is technically adjacent to tailoring nitrile-rubber properties for low-ACN applications.

Patent 2

- Patent Number: US10385647B2

- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses swellable rubber compositions for downhole packers that combine a nitrile elastomer, butadiene rubber, sulfur curing agent, silica, and powder super absorbent polymer. The formulation is processed and cured into swellable packers designed to expand in brine while maintaining mechanical integrity under high temperature and pressure. - Powder super absorbent polymer loading: about 125 Phr to about 175 Phr — claim 2 - Powder super absorbent polymer loading: about 170 Phr to about 210 Phr — claim 3 - Powder super absorbent polymer loading: at least about 150 Phr, about 175 Phr, about 200 Phr or about 225 Phr — claim 4 - Powder SAP fraction of total SAP: at least about 50%, 60%, 70%, 80%, 90%, or 95% by weight — claim 5 - Percent hydrogenation of nitrile rubber elastomer: about 70 percent to about 91 percent — detailed description - Mooney viscosity ML 1+4 (100°C): about 47 to about 72 — detailed description - Percent unsaturation of hydrogenated nitrile elastomer: about 9 percent to about 30 percent — detailed description - Swelling in brine: at least about 50%, 60%, 65%, 70%, 75%, or 80% by volume in about 96 hours — detailed description - Temperature: about 200 degrees Fahrenheit — detailed description and Example 2A/3A - Temperature: about 275 degrees Fahrenheit — detailed description and Example 4A/4B - Pressure: at least about 3000, 4000, 5000 or 6000 psi — detailed description - Pressure: at least about 5000 psi — Example 11A/12A - Brine concentration: about 5% weight/volume — Example 5A - Brine concentration: about 16% weight/volume — Example 6A - Powder SAP loading: 205 Phr — Example 3 and Example 4 - Curing temperature: 330 degrees Fahrenheit — Example 1 - Curing pressure: 70 psi — Example 1 - Curing time: 2.5 hours — Example 1 - Test temperature: 275 degrees Fahrenheit — Example 1 - Test pressure: about 300 psi — Example 1 - Differential pressure resistance: greater than 5000 psi — Example 1 - Base pipe outer diameter: 4.5 inches — Example 1 - Packer outer diameter: about 5.7 inches — Example 1 - Base pipe length: five feet — Example 1 - Hardness: about 86 to 92 Shore A — Example 1 - SAP particle mean size: 35.33 μm — Example 2 - SAP particle mean size: 298.67 μm — Example 2 - SAP particle mean size: 666.23 μm — Example 2 - SAP loading levels: 110 phr, 130 phr, 150 phr, 170 phr — Example 2 - Predicted SAP loading for 75% swell with powder SAP: about 205 Phr — Example 3 - Predicted SAP loading for 75% swell with medium granular SAP: about 190 Phr — Example 3 - Maximum load for medium granular SAP: about 170 Phr — Example 3 - Predicted SAP loading for 75% swell with large granular SAP: about 160 Phr — Example 3 - Maximum load for large granular SAP: about 150 Phr — Example 3

Relevant Experimental Evidence - Example 1: A packer using Composition B was cured at 330°F and 70 psi for 2.5 hours and then withstood >5000 psi differential pressure after exposure to 16% brine at 275°F. - Example 2: Powder, medium granular, and large granular SAPs were

compared at 110-170 phr, showing how particle size and loading affect tensile strength, elongation, hardness, leaching, and swell. - Example 3: Data from Example 2 were used to predict that powder SAP can reach 75% swell at about 205 phr, while granular SAPs are limited by leaching and processability. - Example 4: A 205 phr powder-SAP composition with 10% NBR rubber or 10% HNBR showed excellent green strength and acceptable milling/manipulation properties. - Example 1A: The claims define a sealing method in which the swellable packer is exposed to water-based fluid and heat for 2-5 days to achieve a seal capable of withstanding at least 3000-8000 psi.

Technical Relevance This patent is relevant because it explicitly uses nitrile elastomer-based swellable rubber compositions and provides formulation/process windows for high-temperature, high-pressure service. It does not teach low-ACN synthesis, but it is highly informative for how nitrile rubber is selected and processed in demanding applications where composition tuning matters.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2145681B1 is a nitrile-rubber modification patent focused on metathesis-driven molecular-weight reduction, whereas US10385647B2 is a formulation patent for swellable packers using nitrile elastomer blends.
- EP2145681B1 discloses catalyst-system chemistry with ruthenium/osmium metathesis catalysts and BF₃-adduct activators; US10385647B2 discloses no polymerization catalyst system, instead emphasizing SAP-filled rubber compounding and curing.
- EP2145681B1 provides molecular-weight and Mooney-viscosity context for nitrile rubber processing; US10385647B2 provides Mooney viscosity for a hydrogenated nitrile elastomer (about 47 to about 72) and percent hydrogenation (about 70% to about 91%).
- US10385647B2 contains extensive formulation/process parameters such as 125-175 phr or 170-210 phr powder SAP, 330°F curing, 70 psi, and 2.5 h cure time; EP2145681B1 does not provide comparable formulation data in the extracted evidence.
- Neither patent directly discloses low-acrylonitrile NBR copolymerization or explicit acrylonitrile-content control; both are adjacent technologies that inform property tailoring and downstream use of nitrile rubbers.

4. CONCLUSION

The evidence does not directly disclose a dedicated low-acrylonitrile NBR synthesis route, but it does identify two useful adjacent strategies: metathesis-based molecular-weight reduction of nitrile rubber and high-performance nitrile-elastomer compounding for swellable packers. For low-ACN NBR development, the most actionable synthesis guidance remains outside these patents: control acrylonitrile feed ratio during butadiene/acrylonitrile copolymerization, then tune molecular weight

and Mooney viscosity to the application. Based on the disclosed evidence, lower molecular weight and moderate Mooney viscosity are desirable for processability, while nitrile-elastomer formulations for oilfield service benefit from robust curing and carefully selected SAP loading. Recommended evidence-based targets are a controlled nitrile-rubber feedstock with reduced molecular weight, Mooney viscosity in a processable range, and formulation compatibility with high-temperature brine exposure if the material is intended for swellable applications.

5. REFERENCES

- EP2145681B1 | Method for the reduction of the molecular weight of nitrile rubber | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2145681B1/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>